Where are you? How far did you travel?

LEARNING TARGET I can describe a route using an origin, signed position, total distance, and displacement.

A · KEY VOCABULARY

Position	A signed coordinate in a reference frame: choose an origin and a + direction. $x = -2$ m is 2 m left of the origin.
Distance	Sum of the length of every leg: $\lvert x_2 - x_1\rvert + \lvert x_3 - x_2\rvert + \ldots$ Never negative.
Displacement	$\Delta x = x_f - x_i$. Its sign depends on the axis you chose; the physical motion does not.
Scalar / vector	A scalar has magnitude + units (distance, time). A vector also has direction (displacement).

B · KEY EQUATIONS + UNITS

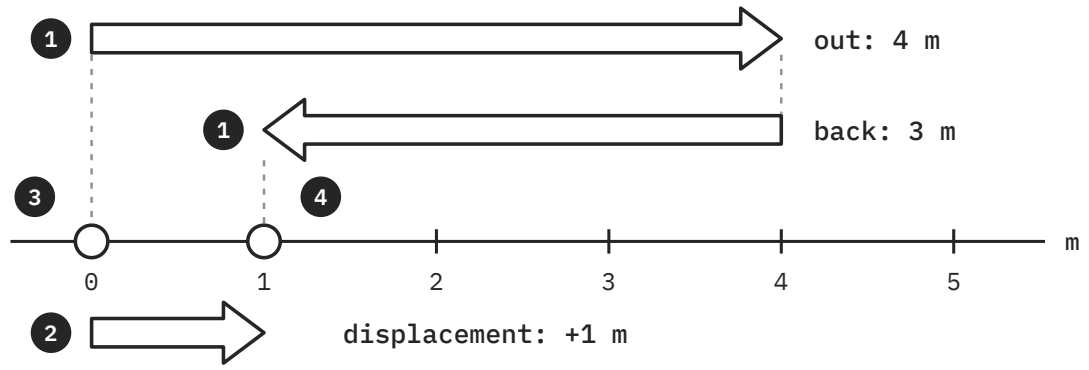
$$\Delta x = x_f - x_i$$

$$\text{distance} = \Sigma\ \lvert\text{each leg}\rvert$$

Δ "change in" (final − initial)
x signed position, meters (m)

C · COLOR IT · ONE TRIP, THREE QUESTIONS

0 → 4 → 1 m: |4 − 0| + |1 − 4| = 7 m; Δx = +1 m



- ☐ **1 · PENCIL** — Distance: trace both legs
- ☐ **2 · BLUE** — Displacement arrow
- ☐ **3 · BLUE** — Start position
- ☐ **4 · BLUE** — Finish position

D · WORKED EXAMPLE

Start at −3 m, move to +2 m, then finish at −1 m. Find the distance and the displacement, with units.

GIVEN	EQUATION	WORK	ANSWER
$x_i = -3$ m turn at +2 m $x_f = -1$ m	distance = $\Sigma\ \lvert\text{leg}\rvert$ $\Delta x = x_f - x_i$	$\lvert 2 - (-3)\rvert + \lvert -1 - 2\rvert = 5 + 3$ $-1 - (-3)$	distance **8 m** displacement **+2 m**

E · THE 2026-XJ THREAD

DAY 2 OF 22 · NEW TOOL: A REFERENCE FRAME

XJ's position only means something in a reference frame: an origin, axes, and a sign convention. Its displacement, not its path length, decides where it ends up.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else